

Survey Data Analysis: A Structured Approach

Techniques, Tools, and Best Practices

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Introduction

- Survey datasets are often large, messy, and heterogeneous.
- Goal: Extract meaningful insights efficiently.
- Importance of a structured approach:
 - Clean the data
 - Explore distributions
 - Identify patterns and inconsistencies
 - Communicate findings clearly

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Exploratory Data Analysis (EDA) Workflow

- 1 Data loading and inspection
- 2 Data cleaning and validation
- 3 Handling missing values
- 4 Categorical variable analysis
- 5 Numerical variable analysis
- 6 Relationships and correlations
- 7 Feature engineering
- 8 Visualisation and reporting

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Python Tools for Survey Data Analysis

- **Pandas:** dataframes, filtering, grouping, reshaping, merging
- **NumPy:** numerical operations and arrays
- **Matplotlib & Seaborn:** visualisations, plots, heatmaps
- **Scipy / Statsmodels:** statistical tests, regressions
- Optional: **Jupyter Notebook** for interactive exploration

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Data Loading and Inspection

- Load datasets using `pd.read_csv()`
- Inspect metadata: `shape`, `columns`, `types`
- Initial null check: `df.isnull().sum()`
- Explore unique values and duplicates

Best Practices

Keep clear, reproducible scripts with modular functions for repeated checks.

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Data Cleaning and Validation

- Handle missing values: imputation or removal
- Correct invalid entries (age ranges, country codes)
- Check consistency across related questions
- Flatten multi-select responses or pivot long-to-wide
- Use reusable functions for common tasks

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Exploratory Analysis

- **Categorical variables:** value counts, distributions, cross-tabs
- **Numerical variables:** mean, median, variance, histograms
- **Relationships:** correlations, scatterplots, heatmaps
- Identify patterns, outliers, and key drivers

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Feature Engineering

- Encode categorical variables (one-hot, binary columns)
- Merge overlapping or related questions
- Derive new metrics from survey responses
- Ensure features are interpretable and relevant

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- Essential for understanding distributions and relationships
- Techniques:
 - Bar charts and histograms
 - Boxplots for outlier detection
 - Heatmaps for correlations
 - Cross-tab plots for multi-variable relationships
- Clear visualisations improve insight communication

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Modular EDA Pipeline

- Organise workflow into reusable functions and scripts
- Example modules:
 - Data loading (`data_utils.py`)
 - Cleaning (`transform_utils.py`)
 - Analysis (`analysis_utils.py`)
 - EDA pipeline runner (`eda_pipeline.py`)
- Config-driven approach for flexibility with new datasets

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Best Practices for Survey Data Analysis

- Maintain reproducible, modular code
- Validate data at every step
- Document assumptions and transformations
- Use visualisation to support conclusions
- Apply statistical reasoning logically

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Conclusion

- Structured approach ensures efficiency and clarity
- Python (Pandas, Matplotlib, Seaborn) provides flexible tools
- Modular pipelines simplify EDA for large survey datasets
- Visualisation and statistical analysis are key to actionable insights

Takeaway

A disciplined, reproducible, and modular EDA workflow turns messy survey data into meaningful, high-quality insights.